

RF-100 Policy Brief No. 1 — AI's Two-Person Rule

A verified quorum protocol for irreversible AI decisions

Remnant Fieldworks Inc. · Proof Before Power™

Companion to **RF-100 — Verified Execution Governance Standard (v1.0 Public Review Draft)**

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Status. This is a public policy brief, not a standard, not certification, and not legal or compliance advice. It summarizes and applies requirements from the RF-100 v1.0 Public Review Draft. RF-100 is itself a draft open for review; no certification program exists yet, and no deployed system — including ExecutionProof™ — is claimed to conform to RF-100 at this maturity level. Independent review findings have been incorporated into RF-100; that is not endorsement.

The problem in one sentence

An AI agent that can take an irreversible action — move money, grant access, delete data, deploy code, sign a contract — can, in most systems deployed today, also be the only party that authorizes that action. The proposer and the approver are the same component. That is exactly the arrangement every mature high-consequence field has spent a century learning to forbid.

The rule humans already trust

High-consequence human systems do not rely on one actor's judgment for irreversible acts. They require an independent second:

- **Nuclear release** requires two authenticated operators acting together — no single person can launch.
- **Wire transfers and treasury movements** above a threshold require dual control or maker-checker separation.
- **Production “break-glass” access** in mature cloud operations requires a second approver and generates a tamper-evident record.

The common principle is the **two-person rule**: for an action whose consequences cannot be undone, authority must be split across independent parties, and the record of who authorized what must survive the action.

AI agents have quietly removed this control. A single model, or a chain of agents all descended from the same model, can propose and approve its own irreversible action. RF-100 restores the two-person rule for machines.

What RF-100 requires

RF-100 makes the two-person rule normative for high-consequence AI execution through four connected requirements:

- **No self-authorization.** “Agents MUST NOT hold, issue, or approve their own authority; authority binding is external to the agent.” (RF-100:AI.9)
- **A real second approver — not a mirror.** For actions at or above the declared consequence threshold, “an M-of-N approver set composed of AI Actors MUST include at least one human approver or at least two approvers of distinct base-model lineage. Approvers sharing base-model lineage SHALL count as one approver for independence purposes.” (RF-100:AI.11)
Two copies of the same model are **one** approver, not two — because they share the same blind spots, the same training, and the same failure modes.
- **Independent authentication, recorded.** “Where M-of-N approval is required, approvals MUST be independently authenticated and recorded in the VDR.” (RF-100.6.2)
- **No borrowed approval down the chain.** In multi-agent workflows, “each consequential hop ... is independently a Governed Action ... An ALLOW rendered for one hop MUST NOT authorize any subsequent hop.” (RF-100:AI.8a) An agent cannot launder a single approval into a sequence of irreversible acts.

Every decision resolves to exactly one of **ALLOW / HOLD / DENY**, and each carries a **ProofRecord** — a tamper-evident receipt of who and what authorized the action, generated before the action executes, not reconstructed afterward. If the quorum is not satisfied, the system **fails closed**: absence of a valid decision means no execution (RF-100.7.2).

Why “distinct lineage” is the whole point

The naive fix — “just add a second AI approver” — fails if both approvers are the same model. Shared base-model lineage means shared training data, shared biases, and shared vulnerability to the same adversarial input or prompt injection. A prompt that fools the proposer will, with high probability, fool its twin.

RF-100:AI.11 closes this by counting approvers who share lineage as **one**. Genuine independence requires either a human in the set or approvers built on **distinct base models**. This converts “two-person rule” from a checkbox into a real diversity-of-failure requirement.

Evidence: this has been tested, not just asserted

Remnant Fieldworks builds standards against preregistered, falsifiable experiments with fixed pass/fail criteria and preserved results. Two are directly relevant to this brief:

Experiment	What it tested	Result
ARK-443	Two-of-three quorum authorization; single compromised approval channel	PASS — quorum enforced; the compromised single channel was denied
ARK-496	Multi-agent delegation and self-approval defense	PASS (8/8) — agents blocked from approving their own delegated authority

These are corpus records, not marketing claims: criteria were set before the runs, and outcomes (including any failures elsewhere in the corpus) are preserved and independently verifiable. See the RF public evidence record and the archived Zenodo depositions for the full corpus and its honest counts.

What this brief is not

- It is **not** a claim that any product today is certified or fully conformant. ExecutionProof implements and internally verifies components of this control story; formal RF-100 conformance remains pending independent external review.
- It is **not** a claim that a second approver eliminates risk. It reduces correlated failure and removes single-actor authority for irreversible acts.
- It is **not** legal advice or a regulatory mandate. It is a proposed control pattern and the normative language RF-100 uses to specify it.

Recommended action for operators

If your AI agents can take **any** irreversible action, ask three questions:

1. **Can the agent approve its own high-consequence action?** If yes, you have no two-person rule. (RF-100:AI.9)
2. **If a second approver exists, is it a different model — or the same model twice?** Same lineage is one approver, not two. (RF-100:AI.11)
3. **Is there a tamper-evident record of who authorized what, created before execution?** If the record is reconstructed after the fact, it is not proof. (RF-100.6.2, Section 8)

Start with **one irreversible action class** — the payment, the deletion, the access grant that would hurt most if it went wrong — and put a verified quorum in front of it. One execution boundary, proven before it scales.

Read the standard: [RF-100 v1.0 Public Review Draft](#) (../RF-100-v1.0-draft.pdf)

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• **Comment:** open a GitHub Issue referencing requirement numbers (e.g., RF-100:AI.11).

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